



NOTE ON TERMINOLOGY

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TUNE computes three output quantities, all within a specified time interval (snapshot):

- Broadband SPL (Sound Pressure Level)
- Decade SPL (" ")
- Broadband PSD (Power Spectral Density)

Names and units used for each quantity are justified as follows.

Broadband SPL

This quantity is obtained by taking the sum of all power spectrum components, expressed as a single broadband value in dB. It is equivalent to taking waveform mean-square in time domain, that is the square of its rms value.

TUNE presents broadband SPL in a plot or tabular form as a series of dB values as a function of time.

Definition is given in ISO 18405:2017 standard as follows:

3.2.1.1

mean-square sound pressure level

root-mean-square sound pressure level

sound pressure level

SPL

$L_{p,rms}$

L_p

DEPRECATED: mean-square SPL

ten times the logarithm to the base 10 of the ratio of the *mean-square sound pressure* (3.1.3.1) to the specified reference value, p_0^2 , in decibels

which in turn derives from:

3.1.3.1

mean-square sound pressure

$\overline{p^2}$

integral over a specified time interval of squared *sound pressure* (3.1.2.1), divided by the duration of the time interval, for a specified frequency range

Symbol and unit for **Broadband SPL** are: $L_{p,rms}$, dB re 1 μPa^2

Note: reference is to 1 μPa^2 since it is based on mean-square sound pressure which is quadratic.

Decidecade SPL

This quantity is obtained by first taking waveform power spectrum, then selecting frequency components that lie inside each decidecade band and adding them together. It is equivalent to filtering in time domain (ideally with infinite frequency rolloff, that is zero transition band), then computing SPL in a similar way as described above for broadband case.

TUNE computes SPL values in a series of decidecade bands and presents them in a plot or tabular form arranged on decidecade center frequency.

Definition is given in ISO 18405:2017 standard as follows, where 'weighted' is meant to be 'decidecade filtered':

3.7.1.3

mean-square weighted sound pressure level

weighted sound pressure level

SPL_w

$L_{p,w}$

ten times the logarithm to the base 10 of the ratio of the *mean-square weighted sound pressure* (3.7.1.1) to the specified reference value, p_0^2 , in decibels

which in turn derives from:

3.7.1.1

weighted sound pressure

p_w

output of a specified linear filter when the input is the *sound pressure* (3.1.2.1), $p(t)$

Symbol and unit for **Decidecade SPL** are: $L_{p,dd}$, dB re 1 μPa^2

where 'dd' is used as a subscript to refer to a decidecade band.

Broadband PSD

This quantity is directly obtained from FFT power spectrum as a series of frequency components starting from zero up to half sampling frequency (Nyquist limit), thus it is named 'Broadband'.

TUNE presents broadband PSD as a series of dB values for each frequency bin: each bin has a fixed width which depends on sampling frequency and number of FFT points.

Definition is given in ISO 18405:2017 standard as follows:

3.2.1.10

mean-square sound pressure spectral density level

$L_{p,f}$

ten times the logarithm to the base 10 of the ratio of the *mean-square sound pressure spectral density*

(3.1.3.13), $\left(\overline{p^2}\right)_f$, to the specified reference value, $\left(\overline{p^2}\right)_{f,0}$, in decibels

which in turn derives from:

3.1.3.13

mean-square sound pressure spectral density

$\left(\overline{p^2}\right)_f$

distribution as a function of non-negative frequency of the *mean-square sound pressure* (3.1.3.1) per unit bandwidth of a sound having a continuous spectrum

Symbol and unit for **Broadband PSD** are: $L_{p,f}$, dB re 1 μPa^2 / Hz